Taylor Bosier

(402) 984-9277 · bosier83@gmail.com · tbosier.github.io

Summary

Lead Data Scientist with 7+ years building production ML, Bayesian inference and optimization models that finance and operations teams run, across warehousing, freight and medical devices. I also build the software around them, from full-stack apps to Rust rewrites of slow pipelines, by directing AI coding agents.

Professional Experience

Lead Data Scientist *Aug 2024 – Present*
Stryker (Remote)

- Built a rebate-accrual forecast for finance (Python, SQL, Databricks, React) that ties out to their workbook. In leakage-safe backtests the shipped hybrid has 0.47x the error of the Excel run rate in use.
- Rebuilt the price-volume-mix revenue bridge that leadership reviews: fixed how fees and rebates were attributed, and showed most of a reported contract decline came from contracts that had ended. Cut load time from 9.8s to 0.6s, with 377 tests that caught all 68 planted bugs.
- Rewrote customer roster matching in Rust: exact matches rose from 20% (legacy SSIS job) to 62%, run time fell 12x and memory 80%. Contract matches are right 93% of the time one is proposed.
- Built a Bayesian hierarchical model of sales rep productivity and sized the surgery-center opportunity from public CMS data to set the payback bar for adding sales reps. Built an invoice-line risk model whose riskiest 1% of lines holds about half of price-credit dollars.
- Moved LLM contract reading to a cheaper model, 26 of 30 contracts fully correct, cutting cost 78% and run time 73%. Built search across the contract estate and chat over a contract with PDF citations.
- Built a causal pipeline for rebate ROI (propensity score matching, inverse propensity weighting) and SKU-level price optimization from isotonic demand models and dynamic programming over price tiers.
- Merged 248 PRs in 12 months across three web apps by directing AI coding agents through written specs and stacked PRs, with tests on 119 of 136 recent PRs. Reviewed 101 teammate PRs, built the team's shared job scheduler, and mentor a senior data scientist and full-stack developers.

Data Scientist II *Sept 2021 – Aug 2024*
Knight-Swift Transportation, Phoenix, AZ *(promoted from Data Scientist I, Sept 2023)*

- Re-architected and deployed a scheduling optimization model (Gurobi) for driver planning, raising planned drivers at one terminal from 20% to 85% and improving on-time delivery and network design.
- Designed an early-warning detection model surfacing 40+ drivers a week for safety remediation.
- Conducted a study using a Bayesian Poisson-Gamma model to quantify the safety impact of lane-keeping sensors, providing critical evidence for a multi-million dollar fleet hardware decision.
- Rebuilt a driver retention model, taking F1 on the churn class from 0.20 to 0.40; the model fed the retention program that ran alongside a 12% fall in driver turnover.
- Built a frequency-severity model of fleet loss exposure that set retention and limits for the annual insurance placement.

Business Analyst *June 2019 – Sept 2021*
Chain Link Services

- Sited warehouses by k-means clustering over demand geography and forecast inventory demand with Prophet, cutting holding costs and stockouts; restructured pick paths with an apriori model and aisle-traversal heuristics, and set offsite levels with a safety-stock rule.

Technical Skills

Modeling and optimization: Python, PyMC, scikit-learn, XGBoost, LightGBM, PyTorch, PySpark, Gurobi, Pyomo
AI and software: LLM agents, RAG, vector search, Claude Code, TypeScript, Next.js, FastAPI, Rust, Docker
Data and cloud: Databricks, Azure, SQL Server, PostgreSQL, Elasticsearch, GitHub Actions, Prefect, Airflow

Open Source

rustmc: Bayesian inference, Python models on a Rust sampler; 4,000+ PyPI downloads since February 2026.

Education

New Mexico State University, Las Cruces, NM: M.S. Mathematics (May 2018), B.S. Mathematics (May 2016)